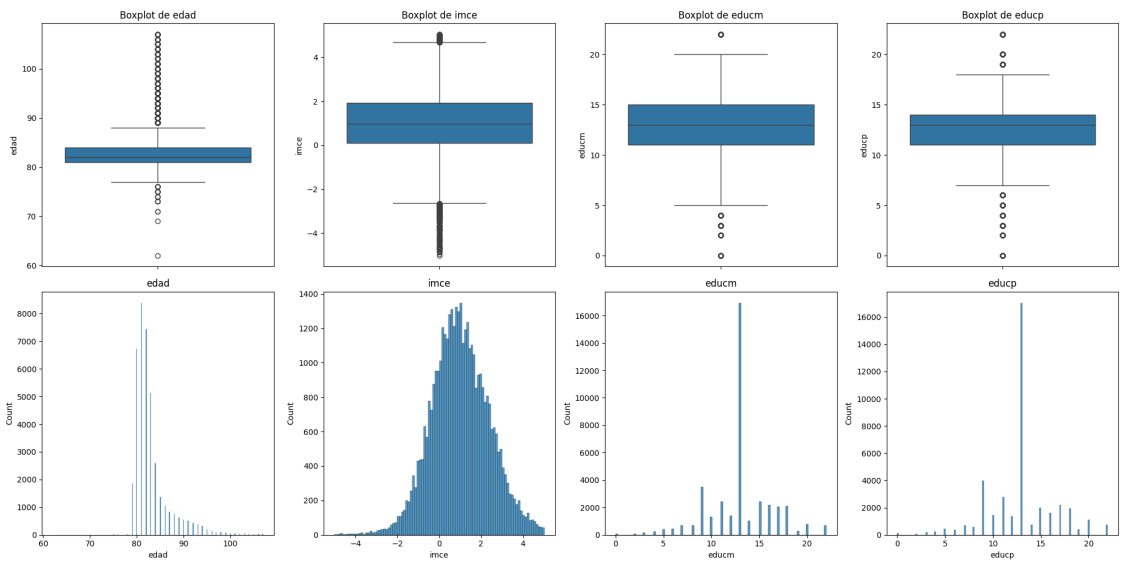
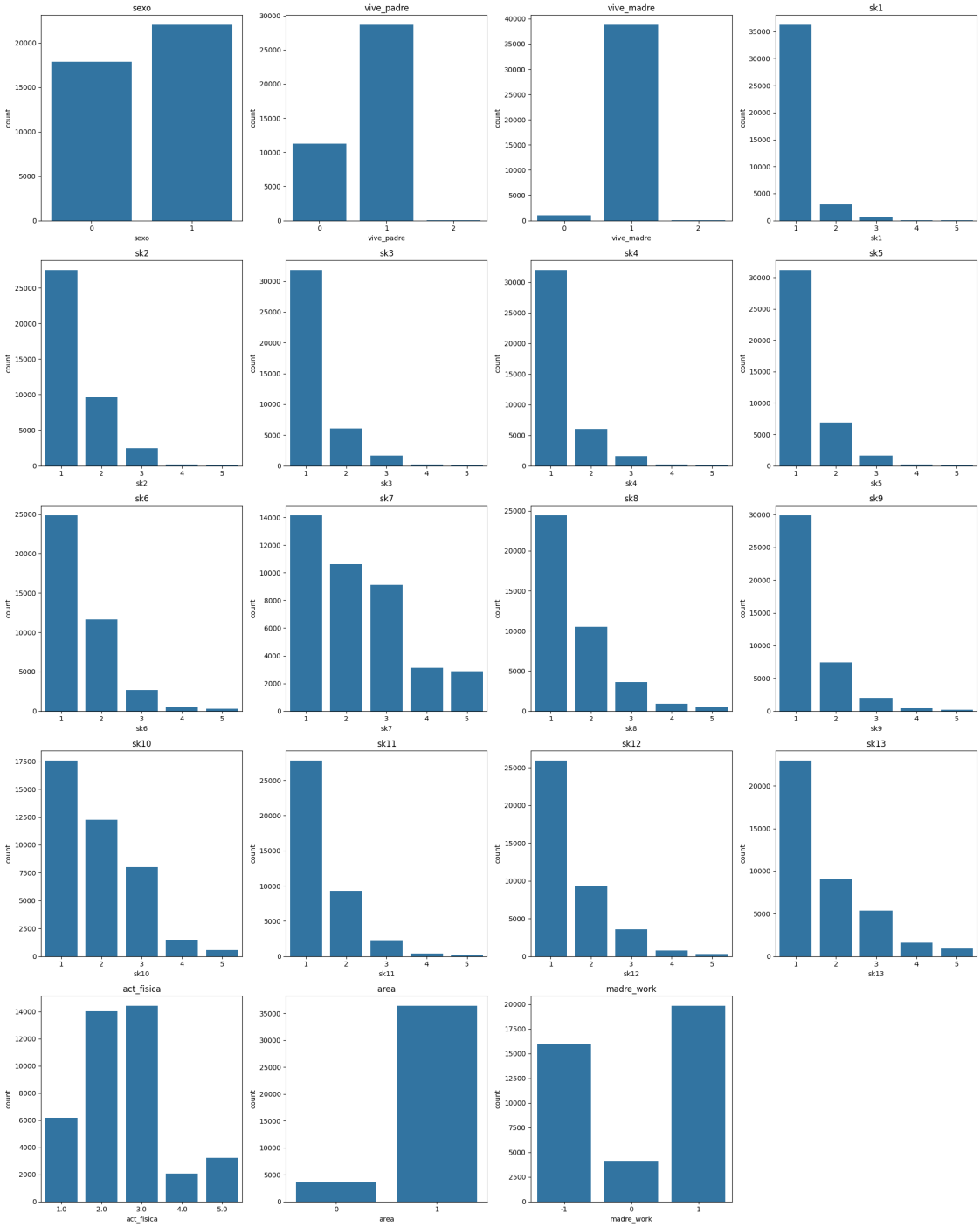
tarea3_PerezMacaya

July 5, 2025

```
       sexo  edad  imce  vive_padre  vive_madre  sk1  sk2  sk3  sk4  sk5  …  \
1         0    76  0.71           0           1    1    1    1    1    1  …
3         1    84  2.05           1           1    1    1    1    1    1  …
4         0    86  1.05           1           1    1    1    1    1    1  …
5         0    74  1.39           1           1    1    2    1    1    1  …
6         1    91  2.75           1           1    1    1    1    2    2  …
…     …   …    …         …          …  …  …  …  …  …  …
41849     0    78  1.63           1           1    1    3    1    2    2  …
41850     1    79  2.57           1           1    1    1    2    2    1  …
41851     0    78  2.12           1           1    1    1    1    1    1  …
41852     1    78 -0.43           1           1    1    1    1    1    2  …
41853     0    78 -0.55           0           1    2    2    1    1    1  …

       sk9  sk10  sk11  sk12  sk13  act_fisica  area  educm  educp  madre_work
1        1     1     1     1     1         5.0     0    8.0      8           1
3        1     1     1     1     1         2.0     1   16.0     12          -1
4        1     1     1     1     1         1.0     1   17.0     15           0
5        1     1     1     1     1         4.0     0    8.0      8          -1
6        3     3     3     2     2         2.0     1   20.0     19           1
…    …   …    …    …   …   …          …   …    …    …          …
41849    2     2     2     1     1         2.0     1   13.0     13          -1
41850    1     3     2     1     4         3.0     1   18.0     19           0
41851    1     3     1     1     1         3.0     1   13.0      9           1
41852    1     2     1     1     2         2.0     1   17.0     15           1
41853    1     1     1     1     1         1.0     1   18.0     11           1

[39898 rows x 23 columns]

sexo            int64
edad            int64
imce          float64
vive_padre      int64
vive_madre      int64
sk1             int64
sk2             int64
sk3             int64
sk4             int64
```

```
sk5             int64
sk6             int64
sk7             int64
sk8             int64
sk9             int64
sk10            int64
sk11            int64
sk12            int64
sk13            int64
act_fisica    float64
area            int64
educm         float64
educp           int64
madre_work      int64
dtype: object
```



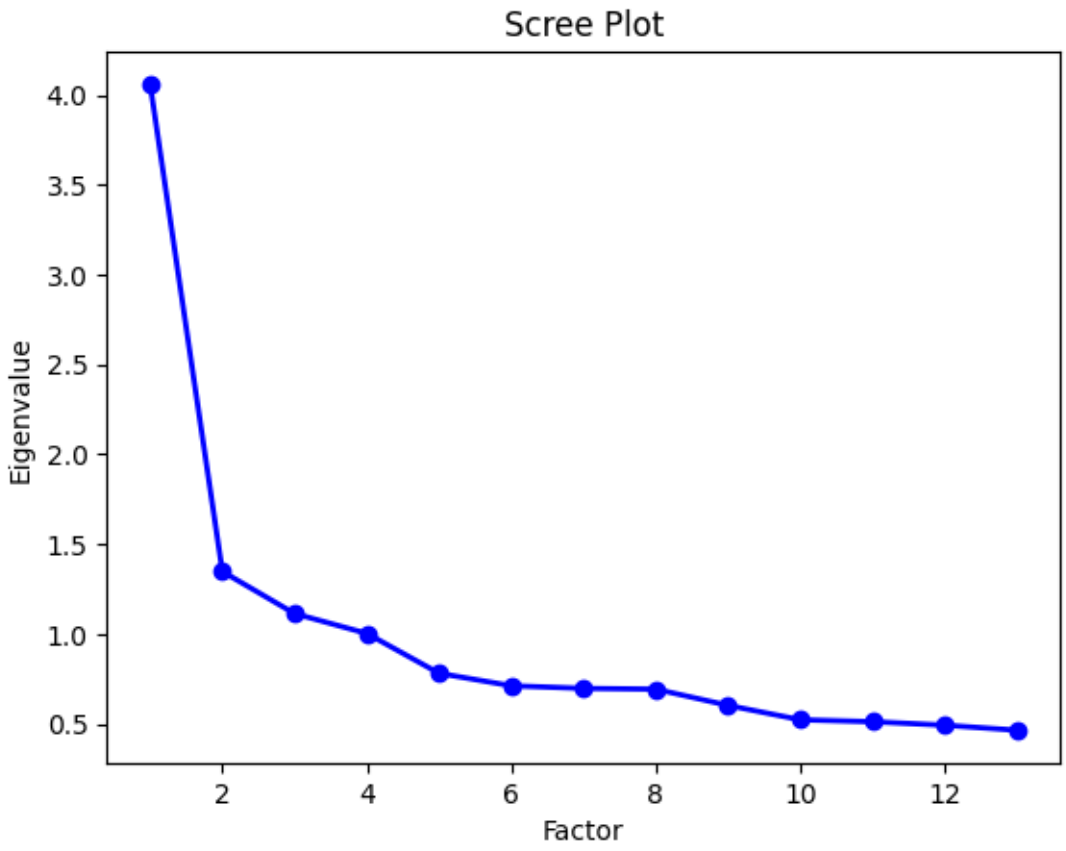


2. EFA

```
c:\Users\edins\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\deprecation.py:132: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

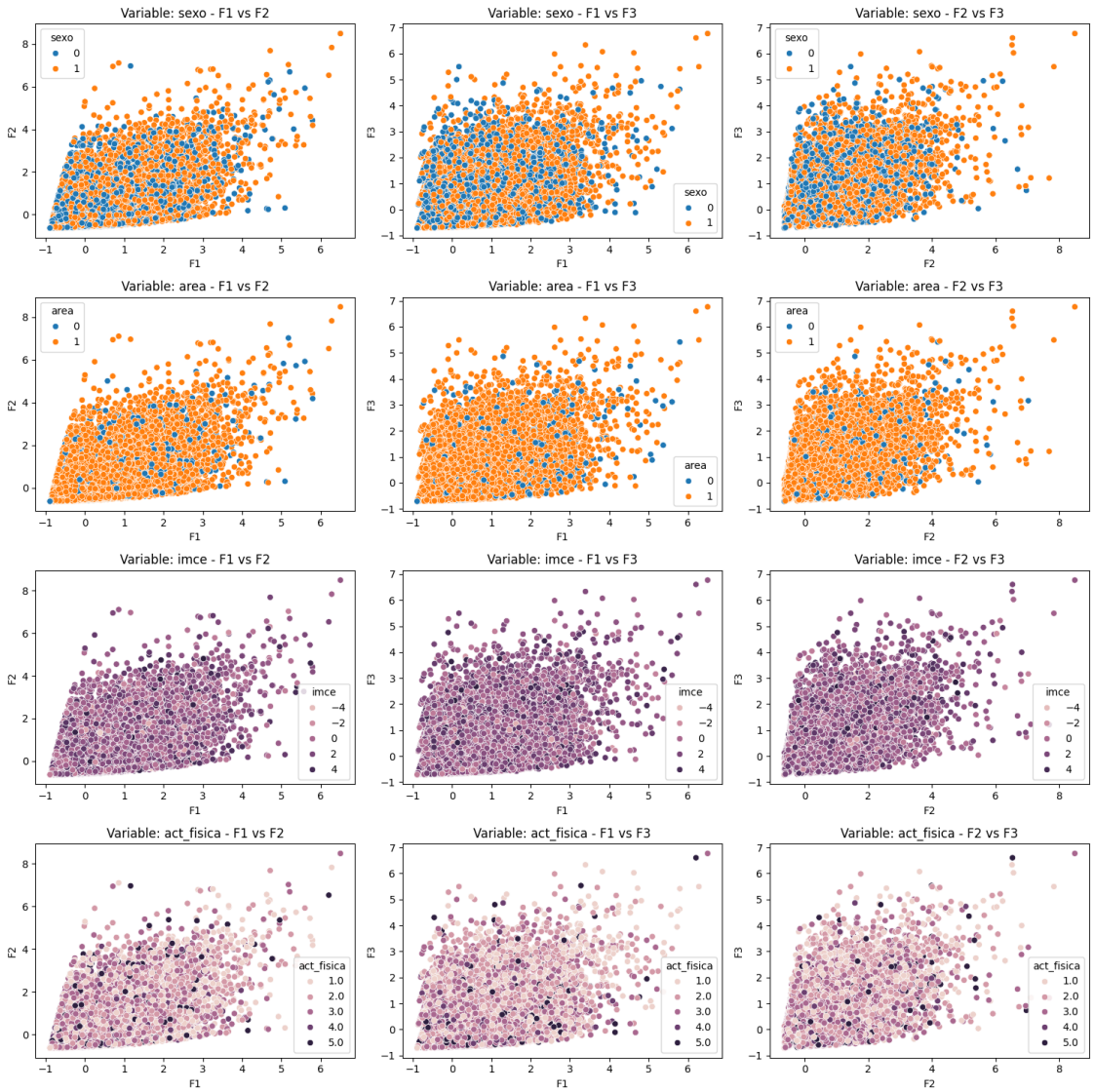
```
array([[ 0.019582,  0.591841, -0.021555],
       [-0.002865,  0.502948,  0.222328],
       [ 0.027641,  0.638366, -0.028415],
       [ 0.006896,  0.730682, -0.013931],
       [-0.033435,  0.023475,  0.772074],
       [ 0.072927,  0.072227,  0.469603],
       [ 0.041492,  0.038748,  0.129864],
       [ 0.214352, -0.079379,  0.46402 ],
       [ 0.502243,  0.079241,  0.052746],
       [ 0.607628, -0.029821, -0.023511],
       [ 0.697418,  0.036714,  0.009946],
       [ 0.5724  , -0.030446, -0.000008],
       [ 0.530523,  0.017536, -0.011748]])
```

Las variables sk1, sk2, sk3 y sk4 están mayormente asociadas al factor 2, mientras que las variables sk5, sk6 y sk8 se encuentran asociadas al factor 3. Por último, las variables desde sk9 hasta sk13 están relacionadas al factor 1. Los valores cercanos a 0 para la variable sk7 indican que no se encuentra mayormente asociada a ningún factor



3.

```
c:\Users\edins\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\deprecation.py:132: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```



Como se puede observar, no se ve ninguna relación clara entre los componentes principales y las variables sexo, area, imce y act_fisica

4. CFA

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.148
```

```
Number of iterations: 38
Params: 1.326 1.234 1.087 1.285 1.877 1.639 1.810 1.017 1.142 0.161 0.054 0.096
0.051 0.060 0.168 0.098 0.620 0.210 0.450 0.726 0.247 0.203 0.167 0.155 0.373
0.484 0.287
```

	lval	op	rval	Estimate	Std. Err	z-value	p-value
0	sk9	~	eta1	1.000000	-	-	-
1	sk10	~	eta1	1.325516	0.015671	84.583493	0.0
2	sk11	~	eta1	1.234209	0.012458	99.069536	0.0
3	sk12	~	eta1	1.087471	0.013103	82.991547	0.0
4	sk13	~	eta1	1.285096	0.016089	79.876675	0.0
5	sk1	~	eta2	1.000000	-	-	-
6	sk2	~	eta2	1.877294	0.020371	92.153655	0.0
7	sk3	~	eta2	1.639420	0.018029	90.934771	0.0
8	sk4	~	eta2	1.810308	0.018777	96.410991	0.0
9	sk5	~	eta3	1.000000	-	-	-
10	sk6	~	eta3	1.017412	0.01216	83.67048	0.0
11	sk8	~	eta3	1.142366	0.01374	83.141749	0.0
12	eta1	~~	eta1	0.161036	0.002781	57.899711	0.0
13	eta1	~~	eta2	0.054415	0.000875	62.215041	0.0
14	eta1	~~	eta3	0.096261	0.001508	63.844859	0.0
15	eta2	~~	eta2	0.051030	0.000902	56.583474	0.0
16	eta2	~~	eta3	0.059945	0.000894	67.077982	0.0
17	eta3	~~	eta3	0.167740	0.002559	65.538574	0.0
18	sk1	~~	sk1	0.097566	0.000813	120.019998	0.0
19	sk10	~~	sk10	0.620184	0.005068	122.374517	0.0
20	sk11	~~	sk11	0.210112	0.002266	92.711786	0.0
21	sk12	~~	sk12	0.450277	0.003638	123.781251	0.0
22	sk13	~~	sk13	0.725986	0.005754	126.163176	0.0
23	sk2	~~	sk2	0.247276	0.002218	111.491836	0.0
24	sk3	~~	sk3	0.203412	0.001789	113.713128	0.0
25	sk4	~~	sk4	0.167309	0.001669	100.242599	0.0
26	sk5	~~	sk5	0.154734	0.001933	80.030393	0.0
27	sk6	~~	sk6	0.372893	0.00324	115.094728	0.0
28	sk8	~~	sk8	0.483673	0.004174	115.871437	0.0
29	sk9	~~	sk9	0.287316	0.002436	117.936087	0.0

Todas las cargas son significativas para sus factores ($p<0.05$). En especial, las variables sk2 y sk4 son las que mas carga muestran en el Factor 2, lo cual representa su alta relación con la expresión de emociones

```
c:\Users\edins\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\deprecation.py:132: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

```
array([[1.      , 0.647804, 0.600301],
       [0.647804, 1.      , 0.585491],
       [0.600301, 0.585491, 1.      ]])
```

Todas las correlaciones son mayores a 0.5, lo cual indica que todos los factores estan moderadamente correlacionados entre si. Esto es lógico tomando en cuenta que todas las variables representan conductas 'positivas'.

5. Stepmix

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[57], line 3
      1␣
 ↪df3=df[['sk1','sk2','sk3','sk4','sk5','sk6','sk8','sk9','sk10','sk11','sk12', sk13','imce']
      2 model = StepMix(n_components=3, measurement="gaussian", verbose=1,␣
 ↪random_state=123)
----> 3 model.fit(df3)
      4 df['pred']=model.predict(df3)

File c:
 ↪\Users\edins\AppData\Local\Programs\Python\Python313\Lib\site-packages\stepmi :\stepmix.
 ↪py:736, in StepMix.fit(self, X, Y, sample_weight, y)
    732     Y = y
    734 if Y is None:
    735     # No structural data. Simply fit the measurement data
--> 736     self.em(X, sample_weight=sample_weight)
    738 elif self.n_steps == 1:
    739     # One-step estimation
    740     # 1) Maximum likelihood with both measurement and structural models
    741     self.em(X, Y, sample_weight=sample_weight)

File c:
 ↪\Users\edins\AppData\Local\Programs\Python\Python313\Lib\site-packages\stepmi :\stepmix.
 ↪py:854, in StepMix.em(self, X, Y, sample_weight, freeze_measurement,␣
 ↪log_emission_pm)
    851 self._check_initial_parameters(X)
    853 # First validate the input and the class attributes
--> 854 X, Y = self._check_x_y(X, Y, reset=True)
    855 n_samples = X.shape[0]
    857 # If sample weights exist, convert them to array (support for lists)
    858 # and check length
    859 # Otherwise set them to 1 for all examples

File c:
 ↪\Users\edins\AppData\Local\Programs\Python\Python313\Lib\site-packages\stepmi :\stepmix.
 ↪py:470, in StepMix._check_x_y(self, X, Y, reset)
    468 if X is not None:
    469     X_names = utils.extract_column_names(X)
--> 470     X = self._validate_data(
    471         X,
    472         dtype=[np.float64, np.float32],
    473         reset=True,
```

```
        force_all_finite=self._force_all_finite_mm,
    )
if Y is not None:
    # Handle 1D Y array
    Y_names = utils.extract_column_names(Y)

AttributeError: 'StepMix' object has no attribute '_validate_data'
```

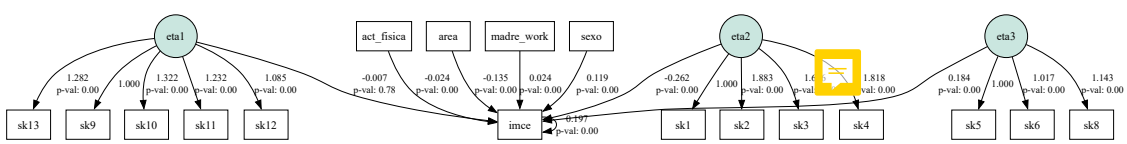
6. SEM

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.303
Number of iterations: 44
Params: 1.322 1.232 1.085 1.282 1.883 1.646 1.818 1.017 1.143 -0.007 -0.262
0.184 0.119 0.197 -0.135 0.024 -0.024 0.162 0.054 0.096 0.051 0.060 0.168 1.219
0.098 0.620 0.210 0.450 0.726 0.247 0.203 0.167 0.155 0.373 0.483 0.287

       DoF  DoF Baseline          chi2  chi2 p-value  chi2 Baseline       CFI  \
Value  117           140  12069.181411           0.0  117112.795319  0.897821

            GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.896944  0.876685  0.896944  0.877734  0.050601  71.394998

              BIC    LogLik
Value  380.781931  0.302501
```

Los valores de CFI y TLI son relativamente altos (~0.9), lo cual indica que se ajustan bien al modelo. Por otro lado, el valor de RMSEA es cercano a 0.05, lo cual también sugiere un buen ajuste

```
WARNING:root:Fisher Information Matrix is not PD.Moore-Penrose inverse will be
used instead of Cholesky decomposition. See 10.1109/TSP.2012.2208105.
```

	lval	op	rval	Estimate	Est. Std	Std. Err	z-value	p-value
0	imce	~	eta1	-0.006919	-0.002017	0.024234	-0.285521	0.775245
1	imce	~	eta2	-0.261755	-0.042733	0.047789	-5.477258	0.0
2	imce	~	eta3	0.183749	0.054555	0.027352	6.717884	0.0
3	imce	~	sexo	0.118831	0.042838	0.011126	10.680941	0.0
4	imce	~	imce	0.196633	0.196633	0.0031	63.434917	0.0
5	imce	~	area	-0.134828	-0.027695	0.019632	-6.867838	0.0
6	imce	~	madre_work	0.024379	0.016643	0.005907	4.126902	0.000037
7	imce	~	act_fisica	-0.023994	-0.018603	0.005191	-4.622518	0.000004
8	sk9	~	eta1	1.000000	0.600182	-	-	-
9	sk10	~	eta1	1.321605	0.559241	0.01562	84.607937	0.0
10	sk11	~	eta1	1.231761	0.733884	0.012418	99.188382	0.0

11	sk12	~	eta1	1.085083	0.545111	0.013065	83.055477	0.0
12	sk13	~	eta1	1.281527	0.517441	0.016039	79.90037	0.0
13	sk1	~	eta2	1.000000	0.584570	-	-	-
14	sk2	~	eta2	1.883176	0.648765	0.020475	91.973847	0.0
15	sk3	~	eta2	1.646199	0.635033	0.018128	90.808982	0.0
16	sk4	~	eta2	1.817584	0.707521	0.018884	96.248809	0.0
17	sk5	~	eta3	1.000000	0.721206	-	-	-
18	sk6	~	eta3	1.016811	0.563267	0.012152	83.674035	0.0
19	sk8	~	eta3	1.143287	0.558571	0.013738	83.218895	0.0
20	eta1	~~	eta1	0.161665	1.000000	0.002787	57.999451	0.0
21	eta1	~~	eta2	0.054348	0.600256	0.000874	62.19029	0.0
22	eta1	~~	eta3	0.096427	0.585601	0.00151	63.876084	0.0
23	eta2	~~	eta2	0.050709	1.000000	0.000899	56.420491	0.0
24	eta2	~~	eta3	0.059719	0.647566	0.000891	66.991046	0.0
25	eta3	~~	eta3	0.167716	1.000000	0.002558	65.559711	0.0
26	imce	~~	imce	1.218930	0.640661	0.006042	201.73718	0.0
27	sk1	~~	sk1	0.097683	0.658278	0.000813	120.185518	0.0
28	sk10	~~	sk10	0.620491	0.687250	0.005068	122.424882	0.0
29	sk11	~~	sk11	0.210137	0.461414	0.002266	92.724792	0.0
30	sk12	~~	sk12	0.450233	0.702854	0.003637	123.790231	0.0
31	sk13	~~	sk13	0.726124	0.732255	0.005754	126.193214	0.0
32	sk2	~~	sk2	0.247427	0.579104	0.002219	111.515421	0.0
33	sk3	~~	sk3	0.203346	0.596733	0.001789	113.652433	0.0
34	sk4	~~	sk4	0.167130	0.499414	0.001669	100.133248	0.0
35	sk5	~~	sk5	0.154729	0.479862	0.001932	80.084799	0.0
36	sk6	~~	sk6	0.373144	0.682730	0.00324	115.176925	0.0
37	sk8	~~	sk8	0.483409	0.687998	0.004173	115.843333	0.0
38	sk9	~~	sk9	0.287132	0.639782	0.002437	117.829549	0.0

WARNING:root:Fisher Information Matrix is not PD.Moore-Penrose inverse will be used instead of Cholesky decomposition. See 10.1109/TSP.2012.2208105.